Factory Method

Season: 1 · Creational · Category: GoF · Level: Beginner

Intent

- Defer instantiation to subclasses (or a switch on a sealed channel).

Problem

- Without this pattern, code tends toward: New things up with conditionals and duplicated construction.

Real-world Analogy

- A restaurant kitchen: the waiter asks for 'dessert'; the pastry chef decides which cake.

When to Use

- Clients must not depend on concrete product classes.
- The design force named in Intent shows up more than once.

When Not to Use

- When a simple constructor or enum switch already suffices.
- A simpler language feature (record, sealed switch, function value) already solves it.

Pattern Structure

- See mermaid diagram in explanation.md

Classic Implementation

```java
package com.buildwithwaqar.patterns.gof.creational.factorymethod;

/** Creator hierarchy — subclasses decide which product to instantiate. */
public abstract class Notifier {
    protected abstract Notification create();

    public final String notify(String to, String body) {
        return create().deliver(to, body);
    }
}

final class EmailNotifier extends Notifier {
    @Override protected Notification create() { return new EmailNotification(); }
}

final class SmsNotifier extends Notifier {
    @Override protected Notification create() { return new SmsNotification(); }
}

final class PushNotifier extends Notifier {
    @Override protected Notification create() { return new PushNotification(); }
}
```

Modern Java 25 / Functional

- NotificationChannel enum with switch replaces Notifier subclasses.

Worked Example

```
See the JUnit test beside the implementation.
Run: mvn -Dtest=*Test test
Focus package matching: gof/creational/factorymethod/
```

Pitfalls

- Don't explode the hierarchy for one-off products.
- Modern Java: sealed enum + switch.

Interview Q&A

- Q: Factory Method vs Abstract Factory? A: Factory Method: one product. Abstract Factory: families of related products.
- Q: How would you test this? A: Prefer injecting interfaces or interpreters; keep demos deterministic (fixed clocks, canned inputs).

Key Takeaways

- Defer instantiation to subclasses (or a switch on a sealed channel).
- Prefer Java 25 sealed types / records / virtual threads where they simplify the classic shape.
- Ship working tests with every episode.

Video Script Highlights

- [0:00] Hook — Open with the analogy: A restaurant kitchen: the waiter asks for 'dessert'; the pastry chef decides which cake. [0:40] Inte...